Ross Program 2026: Application Problems

This document is one part of the application to the Ross Mathematics Program. The full application will be available starting **January 15** at

`https://apply.rossprogram.org/`

Applications must be submitted on or before March 8, 2026 (in your local time). The Admissions Committee will start reading applications after that date.

The Ross Mathematics Program values four qualities in a student above all else: curiosity, exploration, exposition, and rigor. The following three questions are intended to help you develop these qualities. **You are not expected to answer every question completely;** rather, take this application as an opportunity to explore some beautiful mathematics! We are interested in seeing how you approach unfamiliar, open-ended math problems, and we encourage you to write up whatever you discover and think about—especially conjectures or variations of the questions—even if you can't prove them. The Program believes that the most valuable part of a problem is the time spent thinking about it, and your application should reflect this.

Although the Ross Program is highly collaborative, **submit your own individual work on these problems** and do not collaborate with anyone. If you've seen one of the problems before, please include a reference along with your solution. You may use outside non-human resources (e.g., books from the library, a large language model, or other computational tools), but be sure to identify the resources you use and explain your own contributions.

Please write clearly and organize your work so that someone else can follow it: use complete sentences, explain your key steps, and label your diagrams. If you are submitting handwritten work, make sure it is neatly written and scanned or photographed *in good lighting* so that all text and symbols are easy to read. After the application portal becomes available, please upload a PDF containing your work. These PDFs must be at most 10 megabytes in size.

We are not looking for quick answers written in minimal space. Instead, we hope to see organized evidence of your explorations, conjectures, proofs, and generalizations written in a readable format.

Problem 1

For a positive integer n, place $2n$ equally spaced dots around a circle. Those dots are called *nodes*. A line segment connecting two nodes is called an *edge*.

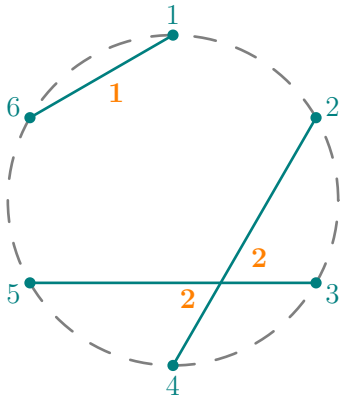
The *circle-distance* between two nodes is the smallest number of steps needed to get from one node to the other. For example, adjacent nodes have circle-distance 1 while opposite nodes have circle-distance n.

The *edge-weight* of an edge is the circle-distance between the two nodes at the ends of that edge.

Draw n edges so that each node is connected to precisely one other node by an edge. Label each of those n edges with its edge-weight.

An *n-card* is such an arrangement of nodes, edges, and edge-weights. Name the $2n$ nodes with the numbers $1, 2, \ldots, 2n$ (in that order) to make discussions clearer.

Here's an illustration of a 3-card, with nodes numbered from 1 to 6 and each edge labeled by its weight.



A *tsuro n-card* is an n-card that has all of its edge-weights distinct. The 3-card pictured above is not tsuro because the weight 2 occurs twice.

(a) For which $n = 1, \ldots, 6$ does there exist a tsuro n-card?

For each case, either provide an example or explain why a tsuro n-card cannot exist.

(b) Are the examples you found in part (a) unique?

How do you define uniqueness in this situation? There may be ways in which two tsuro n-cards look different but are structurally the same.

(c) Can you find examples of tsuro n-cards for some $n > 6$?

(d) **Hard Question:** For which n do tsuro n-cards exist?

Make some conjectures (or guesses). Provide some reasons, proofs or partial proofs for your conjectures.

Problem 2

Here, r, s, m, n are positive integers and k is an integer ≥ 0.

Define the number $r \star s$ recursively by the following rules.

For every r, s, m, k:

(1) $r \star s = s \star r$.

(2) If $r \leq s$ then $r \star m \leq s \star m$.

(3) $2^k \star 2^k = 2^k$.

(4) If $r \leq 2^k$ then $r \star (s + 2^k) = (r \star s) + 2^k$.

(a) Use those rules to prove that $1 \star m = m$ for every m.

Investigate $2 \star n$.

(b) Show that $3 \star 5 = 7$ and $3 \star 6 = 4 \star 5 = 8$.

If $r, s \leq 2^m$ explain why $r \star s \leq 2^m$.

If $2^{m-1} < s \leq 2^m$ show: $2^{m-1} \star s = 2^m$.

For which r, s does it happen that $r \star s = 2^m$?

(c) Prove that $r \star s \leq r + s - 1$ for every r, s.

Find some values of r, s where equality holds: $r \star s = r + s - 1$.

(d) Investigate further properties of $\star$. You might consider one of the following questions, or make up questions of your own:

If $r = 2a$ and $s = 2b$, does $r \star s = 2(a \star b)$? What if $r = 2a - 1$?

If $n \star n = n$, must n be a power of 2 ?

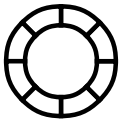
Is the $\star$ operation associative?

Problem 3

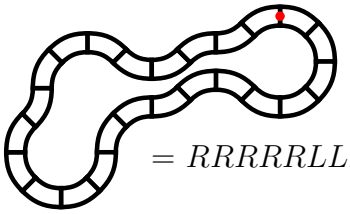
We study "train tracks" built from two kinds of curved pieces:

a left-turn piece (L) and a right-turn piece (R).

Each piece provides a turn by 45°. Then eight left-turn pieces make a complete circle, and so do eight right-turn pieces.



Encode a track by a string in the letters L and R, an "LR-word." To build the track from a word, start at the origin, pointing to the right, and then add pieces in order. (In the pictures, we have not indicated which piece is placed at the origin.)



$= RRRRRLLRLLRRRRRLRRRLLRLR.$

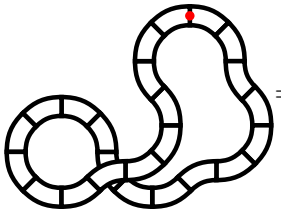
A train track is *closed* if, after placing all pieces, the endpoint equals the starting point *and* the final direction matches the initial direction (so you could connect the pieces together). An LR-word is *closed* if its train track is closed.

(a) How can you quickly tell that $LLLLLLRLRLR$ is not closed?

How can you quickly explain why no train track with 17 pieces can be closed?

(b) **Hard Question:** Given a long LR-word, can you detect whether it is closed?

Here, self-intersections and overlaps are allowed.



$= RRLRRRLRRRLLLLLLLRLLLRRR.$

Find simple conditions on a word $\mathcal{W}$ that guarantee that $\mathcal{W}$ is not closed.

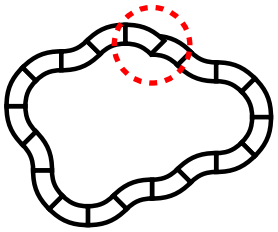
The examples in part (a) hint at such conditions. How about $LRLRLRLR\mathcal{U}$ where $\mathcal{U}$ is an LR-word with only a few letters.

(c) For an LR-word $\mathcal{W}$, create the word $p(\mathcal{W})$ by replacing every L by LLL and every R by RRR. If $\mathcal{W}$ is closed, is $p(\mathcal{W})$ also closed?

Find some other operations that transform closed words to closed words.

(d) Define the *gap* of an LR-word to be the distance between the final endpoint of its track and the starting point.

Among all words that fail to close because the endpoint does not return to the origin, how small can the gap be?



$= RLRRRRLRLRRRLRRRLRR$ is a near miss.

(e) Formulate additional mathematical questions about train tracks. What progress can you make on your questions?